Integrated Gradients: Baselines and Integration Steps

Understanding Feature Contributions to a Model Prediction

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1 What are we trying to explain?

Suppose an already-trained model receives an audio representation x and produces a score $F(x)$ for Parkinson's disease (PD). We may know the prediction, but we would also like to understand which input features influenced it and by how much.

Integrated Gradients (IG) is an attribution method. It explains a particular model prediction relative to a chosen reference input, called the *baseline*. It does not train the model, and it does not attempt to discover the model's entire input-output function. It attributes the difference between the output at the actual input and the output at the baseline.

2 Actual input and baseline representation

Let

$$x = [0.2,\ 0.5,\ 0.8,\ 0.3]$$

be the feature vector representing the actual audio sample. A baseline representation might be

$$x' = [0.1,\ 0.4,\ 0.6,\ 0.2].$$

Here x is the input whose prediction we want to explain, while x' is a reference point against which we compare it.

A baseline can indeed be a vector, but *baseline* does not name one fixed type of vector. Depending on the method, it could be a zero vector, an average feature vector, or another suitable reference input. It does not automatically mean a healthy person's audio; that interpretation is valid only if the method explicitly defines the baseline that way.

3 The path from baseline to actual input

IG constructs intermediate inputs along a straight-line path from x' to x:

$$x(\alpha) = x' + \alpha(x - x'), \qquad 0 \leq \alpha \leq 1.$$

At $\alpha = 0$, we obtain the baseline:

$$x(0) = x'.$$

At $\alpha = 1$, we obtain the actual input:

$$x(1) = x.$$

Values between zero and one give intermediate representations.

For a simple two-dimensional example, take

$$x = [0.8,\ 0.6], \qquad x' = [0,\ 0].$$

Using four equal intervals gives five points:

Step	α	$x(\alpha)$
0	0	[0, 0]
1	0.25	[0.2, 0.15]
2	0.50	[0.4, 0.3]
3	0.75	[0.6, 0.45]
4	1	[0.8, 0.6]

At each intermediate point, IG evaluates the model's gradient with respect to the input features.

4 Why use integration steps?

The model's sensitivity to a feature can change as the input moves from the baseline to the actual sample. A gradient evaluated only at the endpoint gives a local sensitivity at that endpoint; it does not describe how sensitivity behaved along the rest of the path.

Consider the one-dimensional model

$$F(a) = a^2.$$

Move from baseline $a = 0$ to actual input $a = 1$. The output changes from 0 to 1, while the gradient is

$$\frac{dF}{da} = 2a.$$

The gradient is 0 at the baseline and 2 at the actual input. Looking only at either endpoint would fail to represent the full variation in sensitivity along the path. Evaluating gradients at intermediate points lets us approximate the average gradient over the path.

The intuition is similar to travelling from a starting point to a destination: one very large step gives little information about what happened between the endpoints, while many smaller steps examine the path more finely.

5 Integrated Gradients equation

For feature i, the Integrated Gradients attribution is

$$\mathrm{IG}_i(x) = (x_i - x_i') \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} \, d\alpha.$$

This expression has two parts:

- $(x_i - x_i')$ is the difference between the actual feature and its baseline value.
- The integral is the average sensitivity of the model output to feature i along the path from baseline to input.

The integral is generally approximated numerically using a finite number of integration steps. Increasing the number of steps usually makes the numerical approximation finer, at the cost of more gradient evaluations.

6 What does an attribution mean?

Suppose a PD detection model outputs a score such as $F(x) = 0.9$. IG asks how the input features contributed to the change from the baseline output $F(x')$ to the actual output $F(x)$.

For an audio feature vector, dimensions might encode pitch-related information, voice-quality measures, or other acoustic characteristics. IG produces an attribution for each feature. A positive attribution indicates that the feature moves the selected model output upward relative to the baseline; a negative attribution indicates that it moves the output downward. The interpretation depends on which model output or class score F is selected.

The attributions are relative to the selected baseline and the chosen path. They should not automatically be interpreted as causal effects in the real world or as proof that a feature biologically causes Parkinson's disease.

7 Why not just compare the endpoints?

The endpoint difference $F(x) - F(x')$ tells us how much the model output changed overall. It does not, by itself, tell us how to distribute that change across the input dimensions. The gradients along the path provide a principled way to allocate the output change to individual features.

In the ideal continuous formulation, IG satisfies the completeness relation:

$$\sum_i \mathrm{IG}_i(x) = F(x) - F(x'),$$

under the usual differentiability/integration assumptions. A numerical approximation may satisfy this relation only approximately.

8 Choosing the number of integration steps

A small number of steps is computationally cheaper but can yield a rougher approximation. More steps generally improve numerical integration accuracy, although they do not guarantee that the explanation is more meaningful or that the baseline is appropriate. The step count should be checked for convergence by comparing attributions or the completeness residual as the number of steps increases.

9 Connecting the method to the audio-PD example

The process can be summarized as follows:

1. Choose the actual audio representation x whose prediction is to be explained.
2. Choose a meaningful baseline x'.
3. Generate intermediate inputs $x(\alpha)$ between the baseline and actual input.
4. Evaluate gradients of the selected model output at those points.
5. Approximate the path integral and multiply by the feature difference.
6. Interpret the resulting feature attributions relative to the baseline.

10 Final takeaway

Integrated Gradients helps explain a particular prediction by estimating how each input feature contributes to the model-output difference between a baseline and the actual input:

$$\boxed{\text{baseline} \rightarrow \text{intermediate inputs} \rightarrow \text{gradients along the path} \rightarrow \text{feature attributions.}}$$

In short, IG helps us understand which features influenced the model's prediction and by how much, relative to the chosen baseline. It explains an existing model; it does not train the model or establish a complete causal relationship between input and output.